

Aircrack-ng

Category: Wireless Security • 802.11 Analysis • Security Testing

Purpose: Learn the Aircrack-ng suite, understand wireless capture and analysis concepts, and practice authorized Wi-Fi security testing in an isolated lab.

1. What is Aircrack-ng?

Aircrack-ng is a suite of tools for assessing Wi-Fi network security. The suite includes utilities for monitoring wireless traffic, capturing frames, testing authentication mechanisms, and analyzing captured data.

2. Main Tools

Tool	Purpose
airmon-ng	Manage monitor-mode interfaces and identify processes that may interfere with wireless monitoring.
airodump-ng	Capture and display wireless frames and information about nearby access points and stations.
aireplay-ng	Generate selected wireless traffic for authorized testing and lab exercises.
aircrack-ng	Analyze captured authentication material and test keys/passwords against supported capture data.
airdecap-ng	Decrypt supported captured traffic when the required key information is available.

3. Wireless Modes

Normal managed mode is used for ordinary Wi-Fi connectivity. **Monitor mode** allows a compatible wireless interface to observe 802.11 frames without associating in the normal client role. Hardware and driver support vary by adapter and operating system.

4. Check the Wireless Interface

```
iw dev
```

List wireless interfaces on Linux systems.

```
iwconfig
```

Older utility that can display wireless interface information when supported.

```
airmon-ng
```

Display wireless interfaces and monitor-mode related information.

5. Monitor Mode

```
sudo airmon-ng start wlan0
```

Request monitor mode on a compatible wireless interface. The resulting interface name can vary by system and driver; do not assume it will always be wlan0mon.

```
iw dev
```

Use this to confirm the actual interface name after changing modes.

6. Discovering Wireless Networks

```
sudo airodump-ng wlan0mon
```

Display nearby 802.11 access points and stations when the interface is in monitor mode. Use only in an authorized environment.

Important fields commonly shown include BSSID, channel, encryption/authentication information, signal level, packet counts, and associated stations.

7. Capturing Traffic

```
sudo airodump-ng -c 6 --bssid AA:BB:CC:DD:EE:FF -w capture wlan0mon
```

Focus capture on a selected channel and BSSID in a controlled lab. Replace the example values with your authorized lab target.

8. WPA/WPA2 Authentication Material

For WPA/WPA2-Personal testing, a captured four-way handshake contains authentication information that can be used for offline password testing. A capture that shows a client/AP interaction does not automatically mean a usable handshake was

captured.

If aircrack-ng reports that packets contain no EAPOL data, inspect the capture to determine whether the required EAPOL exchange is actually present. A visible client or AP alone is not sufficient.

9. Testing a Captured Key

```
aircrack-ng -w wordlist.txt capture-01.cap
```

Test passwords from a wordlist against supported captured WPA/WPA2 authentication material in your own lab. This is an offline password-testing operation.

10. PMKID vs Handshake

Item	Description
4-way handshake	Authentication exchange between a client and access point; EAPOL frames are involved.
PMKID	An identifier associated with key-management material that may be present in certain captures; it is a different artifact from the four-way handshake.

11. Common Problems

- Wireless adapter does not support monitor mode.
- Driver or firmware issues prevent the interface from behaving correctly.
- Another process interferes with monitor mode.
- The wrong interface name is used after mode changes.
- A capture does not contain the expected EAPOL material.
- The selected channel does not match the target access point.
- The wordlist does not contain the correct password.

12. Defensive Perspective

- Use strong, unique Wi-Fi passwords and modern security protocols such as WPA3 where supported.
- Disable legacy protocols and unnecessary wireless features.
- Monitor for unexpected access points and suspicious association behavior.
- Use enterprise authentication such as 802.1X where appropriate.
- Treat wireless captures as sensitive information.

13. Practical Lab

Create an isolated test network that you own or have explicit permission to assess. Use a compatible USB Wi-Fi adapter, verify monitor-mode support, observe your test AP with airodump-ng, capture normal traffic, and study how authentication frames appear in Wireshark. Keep the exercise disconnected from unrelated networks.

14. Quick Revision

Tool	Remember
airmon-ng	Wireless interface/monitor-mode management.
airodump-ng	Wireless discovery and frame capture.
aireplay-ng	Selected wireless traffic generation for authorized testing.
aircrack-ng	Offline key/password testing against supported captures.
EAPOL	Protocol frames used in WPA/WPA2 authentication exchanges.

15. Key Takeaways

- Aircrack-ng is a suite, not a single program.
- Monitor mode and driver support depend on the wireless adapter.
- A usable WPA/WPA2 handshake requires the relevant authentication frames.

- Practice wireless security testing only on networks you own or are authorized to assess.